



Mark Scheme

Mock 3

Pearson Edexcel GCE
In AS Level Mathematics (8MA0)
Paper 22 Mechanics

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS

General Instructions for Marking

1. The total number of marks for the paper is 30.
2. The Edexcel Mathematics mark schemes use the following types of marks:
 - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
 - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
 - **B** marks are unconditional accuracy marks (independent of M marks)
 - Marks should not be subdivided.

3. Abbreviations

These are some of the traditional marking abbreviations that will appear in the mark schemes.

- bod – benefit of doubt
 - ft – follow through
 - the symbol $\checkmark$ will be used for correct ft
 - cao – correct answer only
 - cso - correct solution only. There must be no errors in this part of the question to obtain this mark
 - isw – ignore subsequent working
 - awrt – answers which round to
 - SC: special case
 - oe – or equivalent (and appropriate)
 - dep – dependent
 - indep – independent
 - dp decimal places
 - sf significant figures
 - * The answer is printed on the paper
 - $\square$ The second mark is dependent on gaining the first mark
4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.
 5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.
 6. Ignore wrong working or incorrect statements following a correct answer.
 7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternatives answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Question		Scheme	Marks	AOs
1(a)		$(T \Rightarrow) 8$	B1	1.1b
			(1)	
(b)		$12a = 15$	M1	3.1b
		$a = 1.25 \text{ (m s}^{-2}\text{)}$	A1	1.1b
			(2)	
(c)		Clear attempt to find the total area under the graph Or use a <i>suvat</i> method for each section and add	M1	2.1
		$\frac{1}{2} \times 15 \times (8 + 30)$ oe	A1	1.1b
		285 (m)	A1	1.1b
			(3)	
(6 marks)				
Notes:				
1(a)	B1	cao		
1(b)	M1	Correct equation in a only		
	A1	cao		
1(c)	M1	Clear attempt with the correct structure (may be implied by correct answer)		
	A1	Correct unsimplified		
	A1	Accept 290 (m)		

Question	Scheme		Marks	AOs
2(a)	Equate sum of i -components to 0: $-1+3+c=0$		M1	3.1a
	$c=-2$		A1	1.1b
			(2)	
2(b)	$(-\mathbf{i}+2\mathbf{j})+(3\mathbf{i}-4\mathbf{j})=0.5\mathbf{a}$		M1	3.1a
	$\mathbf{a}=(4\mathbf{i}-4\mathbf{j})\text{ (ms}^{-2}\text{)}$		A1	1.1b
			(2)	
(4 marks)				
Notes: Allow use of column vectors in working				
2(a)	M1	Equation in c only		
	A1	cao		
2(b)	M1	Use of $\mathbf{F}=m\mathbf{a}$, with all terms		
	A1	Cao. Must be in terms of i and j and not a column vector.		

Question	Scheme		Marks	AOs
3(a)	Differentiate wrt t		M1	1.1b
	$12t + 4 \text{ (m s}^{-2} \text{)}$		A1	1.1b
			(2)	
3(b)	Integrate v wrt t		M1	1.1b
	$x = 2t^3 + 2t^2 + t$		A1	1.1b
	75 (m)		A1	1.1b
			(3)	
(5 marks)				
Notes:				
3(a)	M1	At least two powers decreasing by 1		
	A1	Correct expression		
3(b)	M1	Integrate, with at least two powers increasing by 1 (allow if only two terms integrated)		
	A1	Correct expression		
	A1	cao		

If they use $g = 9.81$ or 10 in this question, penalise once for whole question.

Question		Scheme	Marks	AOs
4(a)		Complete method to find equation in t only: $14.7 - gt = 0$	M1	3.1b
		$t = 1.5$ (s)	A1	1.1b
			(2)	
4(b)		Complete method to find equation in t only	M1	3.1b
		$-19.6 = 14.7t - \frac{1}{2}gt^2$	A1	1.1b
		4 (s)	A1	1.1b
			(3)	
4(c)		e.g. include wind effects, the spin of the stone, size of stone. B0 if any incorrect extras.	B1	3.5c
			(1)	
(6 marks)				
Notes:				
4(a)	M1	Complete method		
	A1	cao		
4(b)	M1	Complete method to find <i>total</i> time. If finding time up and down, need to add them to earn M mark		
	A1	Correct equation(s).		
	A1	Cao. A0 if -1 is not clearly excluded.		
4(c)	B1	e.g. include wind effects, the spin of the stone, size of stone. B0 if any incorrect extras are included		

Question		Scheme	Marks	AOs
5(a)		Equation of motion for A: $T = ma$	B1	3.3
		Equation of motion for B	M1	3.4
		$3mg - T = 3ma$	A1	1.1b
		Solve for T	M1	1.1b
		$T = \frac{3mg}{4} *$	A1*	1.1b
			(5)	
5(b)		$V^2 = 2 \times \frac{3}{4} gh$	M1	2.1
		$V = \sqrt{\frac{3gh}{2}}$	A1	1.b
			(2)	
5(c)		e.g. ignores the mass of the string, ignores stretching of the string. B0 if any incorrect extras.	B1	3.5b
			(1)	
5(d)		V, since air resistance will reduce the acceleration of the particle(s) oe	B1	3.5a
			(1)	
(9 marks)				
Notes:				
5(a)	B1	Correct equation		
	M1	Correct terms and condone sign errors		
	A1	Correct equation		
	M1	Solve for T		
	A1*	Given answer correctly obtained		
		Note: whole system equation could replace one of these equations of motion.		
5(b)	M1	Complete method to find equation in V , g and h		
	A1	Correct expression		
5(c)	B1	B0 if any incorrect extras are given or for an incorrect statement		
5(d)	B1	cao		